AGRIVIVE / KUHL-SPLIT

HIMACHAL PRADESH / KUHL-INSPIRED / TWO-BRANCH GRAVITY DIVIDER

SERIES 09 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



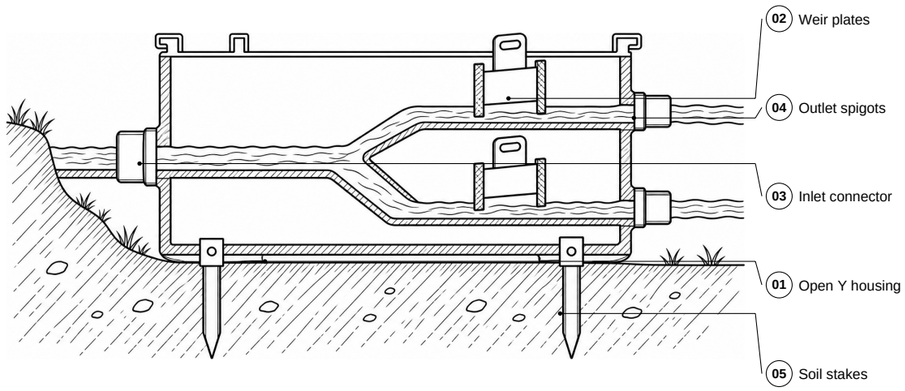
STATE CONTEXT / INDIGENOUS INSPIRATION

Kuhls convey hill-stream water through gravity channels to irrigated fields in Himachal Pradesh. KUHL-SPLIT proposes two independently adjustable branch weirs; flow allocation needs site calibration.

Practice source: CGWB manual traditional recharge systems | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 180 L x 120 W x 90 H

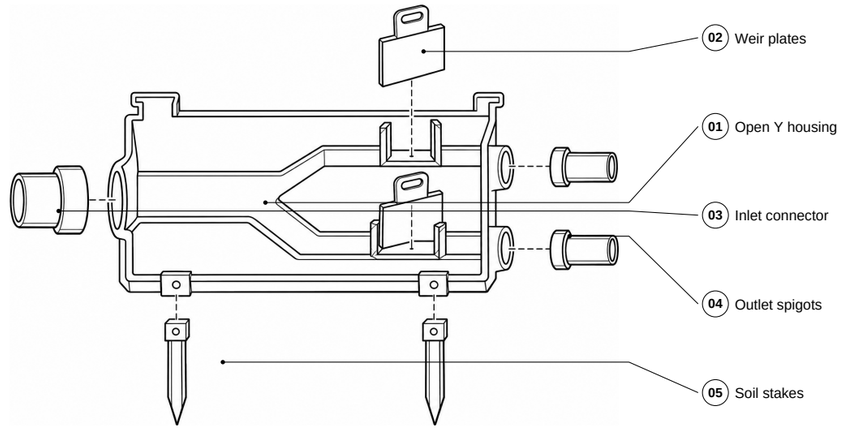


DESIGN / OPERATION NOTES

Level and anchor the open divider at a gravity-channel junction.
Set branch plates independently; provide an unobstructed downstream path.
Inspect and calibrate branch flows; equal outlet areas do not ensure equal delivery.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
180 L x 120 W x 90 H

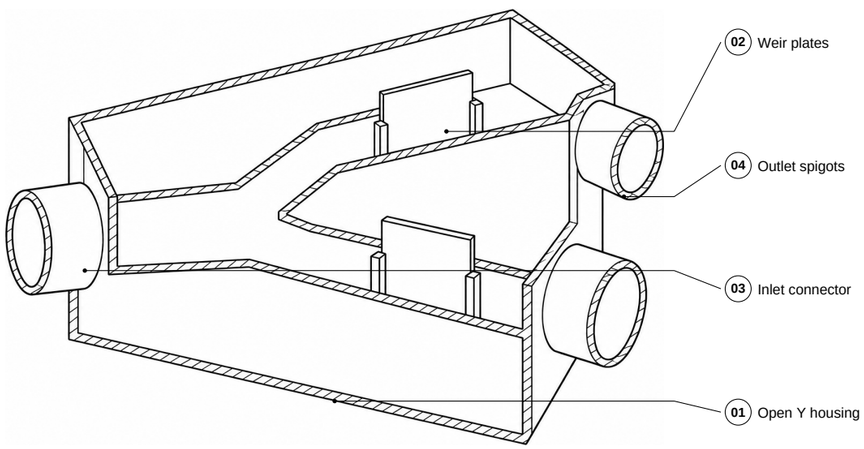


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

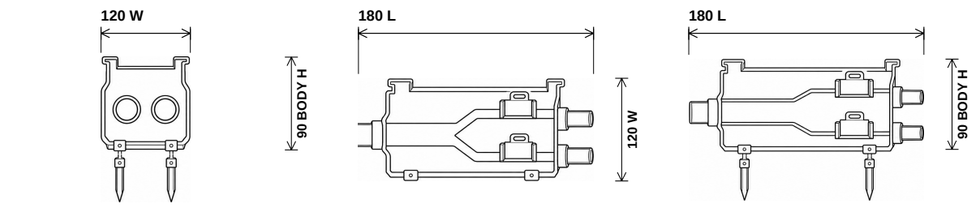
SECTION BASIS / mm
30 inlet ID; two 20 outlet IDs; gate travel TBD



DESIGN / OPERATION NOTES

Set branch plates independently; provide an unobstructed downstream path.
Inspect and calibrate branch flows; equal outlet areas do not ensure equal delivery.

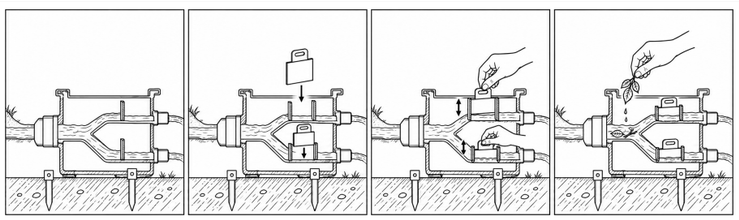
03 ORTHOGRAPHIC STUDIES / mm



OUTLET END **PLAN** **SIDE STUDY**

Housing 180 L x 120 W x 90 H. Inlet DIA 30 ID; two outlets DIA 20 ID. Stake length / gate travel TBD.

05 INSTALLATION / OPERATING SEQUENCE



1 LEVEL / STAKE **2 FIT PLATES** **3 ADJUST BRANCHES** **4 INSPECT / CLEAR**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Housing L x W x H
180 x 120 x 90

Inlet ID
30

Two outlet IDs
20 each

02 GEOMETRIC CHECK

COMBINED OUTLET BORE AREA

628.3 mm2

2 x pi/4 x 20^2
Does not establish equal flow division

03 PARTS / QTY

01 Open Y housing	**1**
02 Weir plates	**2**
03 Inlet connector	**1**
04 Outlet spigots	**2**
05 Soil stakes	**2**

Housing/plates: candidate polymer. Connectors and ground stakes: material/fit TBD.

04 INTERFACES / CHECKS

Check leakage around plates and debris blockage.

Survey inlet and branch invert levels.

Measure flow allocation at observed operating heads.

05 DATUMS / DRAWING BASIS

A: inlet axis. B: housing base. Survey branch inverts before setting plates.
PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-09 / P01	KUHL-SPLIT	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	09 / 28